

Direct recognition

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$v_y = +12$ at $t = 1$ s, $a_y = -10$. Find u_y .

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Place launch and event on one clock.

H2 Use signed vertical constant acceleration.

H3 $12 = u_y - 10$.

CHECK AFTER ATTEMPT - DIRECT RECOGNITION

$u_y = 22$ m/s upward.

This is direct recognition of the taught structure.

Near transfer

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$v_y = +6$ at $t = 2$ s, $a_y = -9$. Find u_y .

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Do not invent a new model because numbers changed.

H2 Keep the same sign and clock.

H3 $6 = u_y - 18$.

CHECK AFTER ATTEMPT - NEAR TRANSFER

$u_y = 24$ m/s upward.

Surface numbers changed; hidden structure did not.

Reversed target

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$u_y=30$, later $v_y=+3$, $a_y=-9$. Find t .

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Target changed; physics did not.

H2 Use the same relation for a different unknown.

H3 $3=30-9t$.

CHECK AFTER ATTEMPT - REVERSED TARGET

$t=3$ s.

Same invariant, different target.

Model selection

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

Which needs this model: horizontal range; infer u_y from later v_y and t ; centripetal acceleration?

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Look for stated vertical velocity at stated time.

H2 This capability is signed one-dimensional vertical velocity under constant acceleration.

H3 The second option matches.

CHECK AFTER ATTEMPT - MODEL SELECTION

The vertical-event option is the match.
Core2B tests recognition and selection, not only algebra.

Exit robustness

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$u_y=36$, $v_y=0$, $a_y=-9$. What event is this and when?

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 What event has $v_y=0$?

H2 Use the same one-clock signed model.

H3 $0=36-9t$.

CHECK AFTER ATTEMPT - EXIT ROBUSTNESS

Apex event; $t=4$ s.

Event recognition + model selection + changed target.